

# LAB 01

## Introduction to Programmable Logic Controllers

**Student Name:** Adamu Nando Usufa

**Registration Number:** C11/26/ACIS/17367

**Course:** Intro to Critical Infrastructure for IT  
Learners

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# 1. Laboratory Setup

The ICDFA PLC virtual machine was configured in VirtualBox using an isolated Internal Network. A clean snapshot named LAB01-START was created before beginning the laboratory.

The PLC VM was configured with the expected laboratory IP address:

192.168.1.10

## 1.1. LAB01-START Snapshot

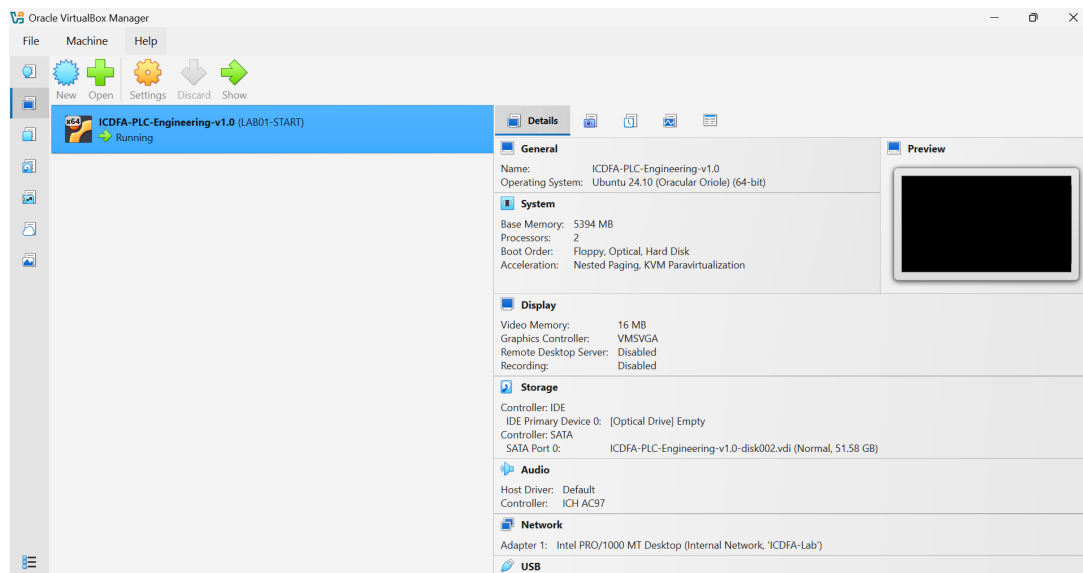


Figure 1: LAB01-START snapshot in VirtualBox.

# 2. IP Address Verification

The PLC VM IP address was verified from the terminal and confirmed as:

192.168.1.10

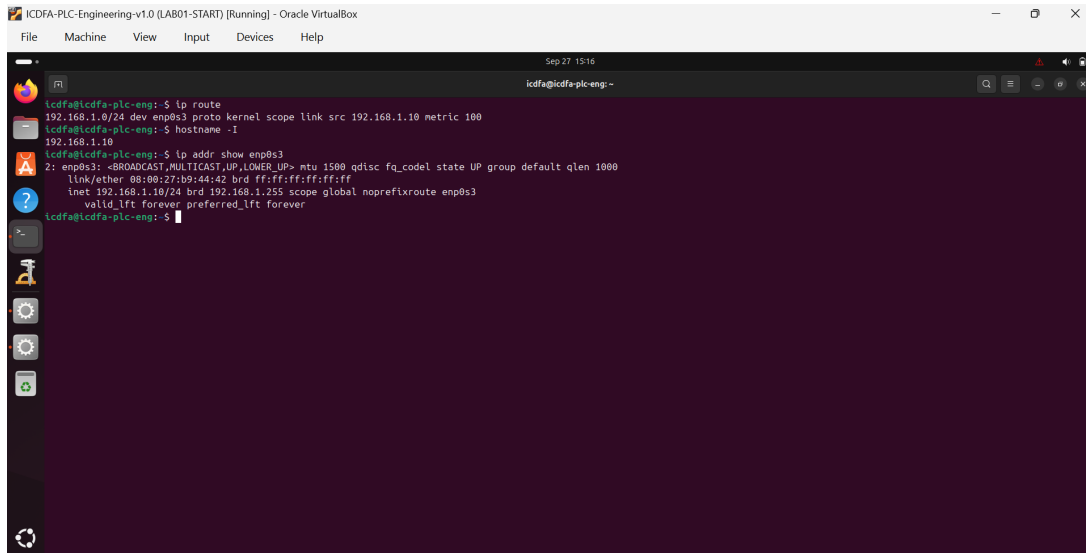


Figure 2: PLC VM IP address verification.

### 3. OpenPLC Motor Control

The OpenPLC Editor was opened and the provided Ladder Diagram project was loaded. Under the project tree, Res0 and instance0 (Motor\_Control) were opened. The diagram was expanded so that the complete motor-control logic was visible.

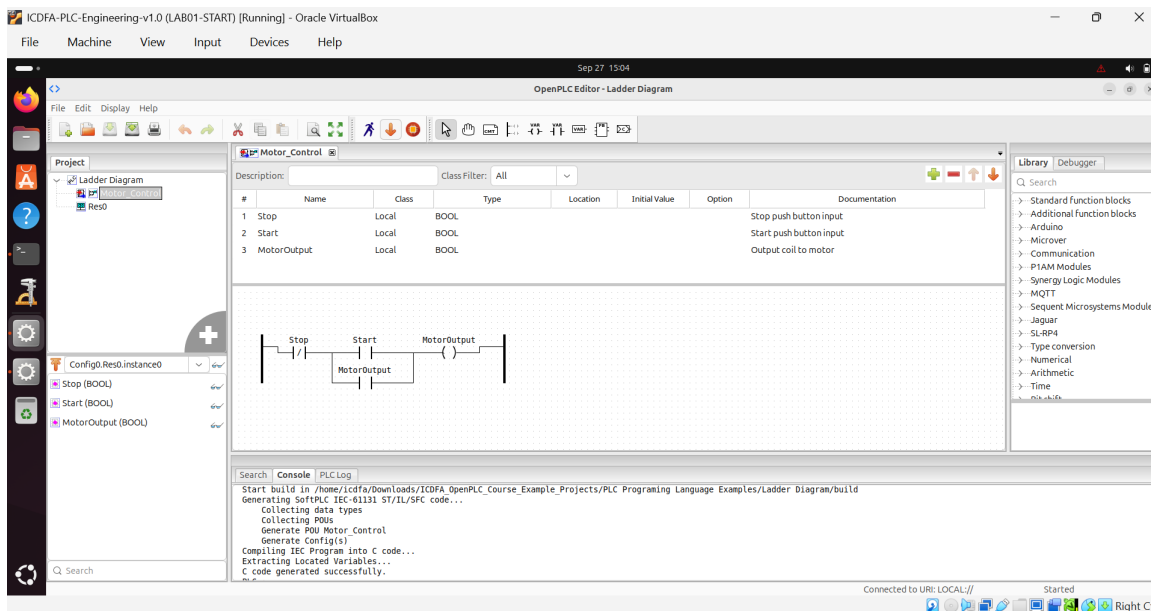


Figure 3: Complete Motor Control ladder diagram in OpenPLC Editor.

## 4. Motor Control Logic

The main elements identified in the ladder diagram are:

| Element                    | Function                                                 |
|----------------------------|----------------------------------------------------------|
| <b>Stop Contact</b>        | Normally closed contact used to stop the motor.          |
| <b>Start Contact</b>       | Normally open contact used to start the motor.           |
| <b>MotorOutput Contact</b> | Holding/latching contact that keeps the motor energized. |
| <b>MotorOutput Coil</b>    | Output coil representing the motor output.               |
| <b>Power Rails</b>         | Provide the ladder logic path.                           |

The motor output is energized when the Stop path is closed and the Start contact is activated. Once energized, the MotorOutput contact maintains the circuit after the Start contact is released.

The motor output is de-energized when the Stop contact opens and breaks the control path.

The logic can be represented as:

$$\text{MotorOutput} = \text{NOT Stop} \wedge (\text{Start} \vee \text{MotorOutput})$$

## 5. I/O Classification

| I/O Type               | Example              | Description                               |
|------------------------|----------------------|-------------------------------------------|
| <b>Discrete Input</b>  | Start push button    | ON/OFF control signal.                    |
| <b>Analog Input</b>    | Temperature sensor   | Continuous temperature measurement.       |
| <b>Discrete Output</b> | Motor ON/OFF         | Turns a motor on or off.                  |
| <b>Analog Output</b>   | Motor speed setpoint | Continuous control value for motor speed. |

## **6. Knowledge Check**

### **1. Difference between an input module and an output module**

An input module receives signals from field devices such as switches and sensors. An output module sends control signals from the PLC to field devices such as motors and valves.

### **2. Why is a normally closed contact commonly used for stop or emergency-stop logic?**

A normally closed contact opens when the stop condition is activated, breaking the control path and causing the output to de-energize.

### **3. Give one discrete I/O example and one analog I/O example**

A Start push button is an example of a discrete input. A temperature sensor is an example of an analog input.

### **4. What operational risk can arise when industrial automation devices become network-connected?**

Network-connected automation devices can become exposed to unauthorized access or malicious network activity, which may disrupt industrial operations.

## **7. Reflection**

Network-connected PLCs provide benefits such as remote monitoring, communication with other industrial systems and easier maintenance. However, network connectivity also introduces cybersecurity risks. Unauthorized access, malicious traffic or a compromised connected system could affect the operation of a PLC and potentially disrupt an industrial process. Therefore, PLCs should be protected using appropriate network security and access controls while maintaining the availability and operational requirements of the industrial environment.